WORKED SOLUTION

STEP III 2023, Question 2

Pure • Polar Curve Areas

STEP III 2023, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Both curves are drawn by a radius sweeping from $\theta = 0$ to $\theta = \pi$, so the whole question is about which curve is further out at each angle. Comparing $k(1 + \sin\theta)$ with $k + \cos\theta$ shows that they swap over exactly once, at $\theta = \alpha$. Regions A and B together are then the overlap of the two curves, and every area comes from the polar formula $\frac{1}{2}\int r^2 \, \mathrm{d}\theta$ with $\sin^2$ and $\cos^2$ handled by double angles.

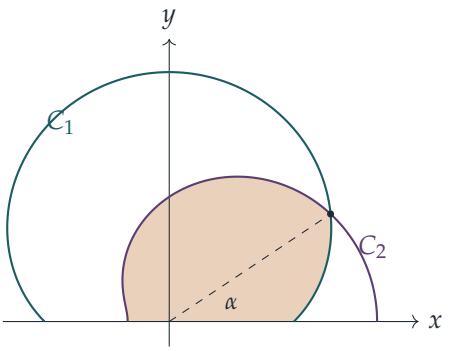
Solution

Part (i)

Some values help with the sketch:

	$\theta = 0$	$\theta = \frac{\pi}{2}$	$\theta = \pi$
C_1: $r = k(1 + \sin\theta)$	k	$2k$	k
C_2: $r = k + \cos\theta$	$k + 1$	k	$k - 1$

Both radii are positive throughout (as $k > 1$). C_1 is the upper half of a cardioid, symmetric about the y-axis and reaching $(0, 2k)$. C_2 is a slightly squashed semicircle of "radius about k", pushed 1 unit to the right at $\theta = 0$ and 1 unit in at $\theta = \pi$. The sketch below has $k = 1.5$; the shaded region is where the curves overlap.



At an intersection the radii agree, so

$$k(1+\sin\alpha) = k+\cos\alpha \iff k\sin\alpha = \cos\alpha.$$

Here $\cos\alpha \neq 0$ (if it were, we would need $k\sin\alpha = 0$, but $\sin\alpha = \pm 1$ then), so dividing gives $\tan\alpha = \dfrac{1}{k}$, as required. Since $0 < \frac{1}{k} < 1$ there is exactly one such angle in $[0,\pi]$, with $0 < \alpha < \frac{\pi}{4}$.

More usefully, $k+\cos\theta - k(1+\sin\theta) = \cos\theta - k\sin\theta$, which is positive for $0 \leq \theta < \alpha$ and negative for $\alpha < \theta \leq \pi$ (using the values of $\sin\alpha$, $\cos\alpha$ below, it equals $\sqrt{1+k^2}\,\sin(\alpha-\theta)$, and $\alpha-\theta$ lies in $(0,\alpha]$ or in $[\alpha-\pi, 0)$ respectively). So C_2 is outside C_1 before α, and inside it afterwards.

We will also use, from $\tan\alpha = \frac{1}{k}$,

$$\sin\alpha = \frac{1}{\sqrt{1+k^2}}, \qquad \cos\alpha = \frac{k}{\sqrt{1+k^2}}.$$

Part (ii)

Region A is swept out by the radius up to C_1 for $0 \leq \theta \leq \alpha$, so its area is

$$\begin{aligned}
\tfrac{1}{2}\int_0^\alpha k^2(1+\sin\theta)^2\,\mathrm{d}\theta &= \frac{k^2}{2}\int_0^\alpha \left(1+2\sin\theta+\tfrac{1}{2}-\tfrac{1}{2}\cos 2\theta\right)\mathrm{d}\theta \\
&= \frac{k^2}{2}\left[\tfrac{3}{2}\theta - 2\cos\theta - \tfrac{1}{4}\sin 2\theta\right]_0^\alpha \\
&= \frac{k^2}{2}\left(\tfrac{3}{2}\alpha - \tfrac{1}{2}\sin\alpha\cos\alpha - 2\cos\alpha + 2\right) \\
&= \frac{k^2}{4}(3\alpha - \sin\alpha\cos\alpha) + k^2(1-\cos\alpha),
\end{aligned}$$

using $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$ and $\sin 2\alpha = 2\sin\alpha\cos\alpha$. This is the required expression.

Part (iii)

Similarly, using $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$, the area of B is

$$\begin{aligned}\tfrac{1}{2}\int_\alpha^\pi (k+\cos\theta)^2\,\mathrm{d}\theta &= \tfrac{1}{2}\int_\alpha^\pi \left(k^2+\tfrac{1}{2}+2k\cos\theta+\tfrac{1}{2}\cos 2\theta\right)\mathrm{d}\theta \\ &= \tfrac{1}{2}\left[\left(k^2+\tfrac{1}{2}\right)\theta + 2k\sin\theta + \tfrac{1}{4}\sin 2\theta\right]_\alpha^\pi \\ &= \frac{(2k^2+1)(\pi-\alpha)}{4} - k\sin\alpha - \tfrac{1}{4}\sin\alpha\cos\alpha.\end{aligned}$$

If we want to remove the trigonometric functions of α, the values from part (i) give $k\sin\alpha = \dfrac{k}{\sqrt{1+k^2}}$ and $\sin\alpha\cos\alpha = \dfrac{k}{1+k^2}$, so

$$\text{area of B} = \frac{(2k^2+1)(\pi-\alpha)}{4} - \frac{k}{\sqrt{1+k^2}} - \frac{k}{4(1+k^2)}.$$

Part (iv)

Ask yourself What shape is the region made of A and B together?

By part (i), for $\theta < \alpha$ the boundary of A is the inner curve C_1, and for $\theta > \alpha$ the boundary of B is the inner curve C_2. So R is the area of the overlap of the two regions (shaded in the sketch). That already suggests the answers: for large k, C_2 lies almost entirely inside C_1, so the overlap should be almost all of T.

The full areas are, by the same integrals taken over $[0,\pi]$ (the $\sin\theta$ and $\sin 2\theta$ terms give 0 at both ends; $-2\cos\theta$ gives 4):

$$T = \tfrac{1}{2}\int_0^\pi (k+\cos\theta)^2\,\mathrm{d}\theta = \frac{\pi(2k^2+1)}{4}, \qquad S = \tfrac{1}{2}\int_0^\pi k^2(1+\sin\theta)^2\,\mathrm{d}\theta = \frac{k^2(3\pi+8)}{4}.$$

As $k\to\infty$, $\alpha = \arctan\frac{1}{k} \to 0$. Rather than expand A, we bound it: on $[0,\alpha]$ we have $r = k(1+\sin\theta) \le 2k$, so A lies inside a sector of radius $2k$ and angle α, and since $\alpha < \tan\alpha = \frac{1}{k}$,

$$0 < \text{area of A} \le \tfrac{1}{2}(2k)^2\alpha < 2k.$$

Hence $\dfrac{\mathrm{A}}{T} < \dfrac{8k}{\pi(2k^2+1)} \to 0$. For B, dividing part (iii) by T,

$$\frac{\mathrm{B}}{T} = \frac{\pi-\alpha}{\pi} - \frac{4\left(k\sin\alpha + \frac{1}{4}\sin\alpha\cos\alpha\right)}{\pi(2k^2+1)} \to 1 - 0 = 1,$$

because $\alpha\to 0$ and $0 < k\sin\alpha + \frac{1}{4}\sin\alpha\cos\alpha < k+1$, which is small compared with $2k^2+1$. Therefore

$$\frac{R}{T} = \frac{\mathrm{A}}{T} + \frac{\mathrm{B}}{T} \to 1.$$

Finally

$$\frac{R}{S} = \frac{R}{T}\cdot\frac{T}{S} = \frac{R}{T}\cdot\frac{\pi(2k^2+1)}{k^2(3\pi+8)} \to 1\cdot\frac{2\pi}{3\pi+8} = \frac{2\pi}{3\pi+8}.$$

Checks

- At $\theta = 0$, C_2 ($r = k+1$) is outside C_1 ($r = k$), and at $\theta = \frac{\pi}{2}$ it is inside ($k < 2k$), so the curves must cross between, consistent with $0 < \alpha < \frac{\pi}{4}$.
- With $k = 2$: $\alpha = \arctan\frac{1}{2} \approx 0.4636$, A ≈ 1.4132, B ≈ 5.0309, $T = \frac{9}{4}\pi \approx 7.0686$, so $R/T \approx 0.912$. Direct numerical integration of $\frac{1}{2}r^2$ agrees with both formulas.
- For $k = 50$, $R/T \approx 0.99987$ and $R/S \approx 0.36061$, close to $\frac{2\pi}{3\pi+8} \approx 0.36059$.
- A sanity check on the limit: for huge k, C_2 is nearly a semicircle of radius k (area $\frac{1}{2}\pi k^2$), and C_1 is k times a fixed half-cardioid of area $\frac{3\pi}{4} + 2$; the ratio is $\frac{2\pi}{3\pi+8}$.

Common slips

- Forgetting the factor $\frac{1}{2}$ in the polar area formula, or squaring $1 + \sin\theta$ as $1 + \sin^2\theta$.
- In part (iv), arguing that $k^2(1 - \cos\alpha)$ "tends to 0 because $\alpha \to 0$": it is multiplied by k^2, and in fact it tends to $\frac{1}{2}$. Only its size compared with T matters.

Beyond STEP

The area of the overlap of two regions given in polar form is a first taste of the area integrals $\iint r\,\mathrm{d}r\,\mathrm{d}\theta$ met in first-year multivariable calculus, where the hard part is always describing the region, exactly as here. The limit arguments in part (iv) (bounding a term rather than computing it exactly) are the style of real analysis; Abbott's *Understanding Analysis* is a good place to see that style developed carefully.

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